A1 Week 2 Plan

General Carbon/Contracting Theory & Numerical Methods — closing the gap to a *Management Science* submission

Status. Week 1 delivered the spine: the capacity-substitution channel $K_R^* > K_R^\circ$, the exact net-carbon inequality (C), the distribution-free cutoffs, the closed-form thresholds $(\bar{C}_M, C_R^\dagger, C_R^\circ, \bar{C}_M^b)$, a 15/15 machine-precision audit, a 3-referee honesty pass, and a green four-role integration. An adversarial *Management Science* review puts us at ≈70–75% of the rigor bar. The gap is one keystone — a general-distribution, finite-optimum carbon theorem — plus robustness, a clean numerical narrative, and theory positioning.

Scope (hard wall — A1 only). Every task below is general *theory* or *numerical-method* work on the FCL carbon model. *Out of scope:* real-data calibration/magnitudes (A2); reliability, welfare, social-vs-private optimum (B1); flexibility continuum, AI, storage, multi-grid, dynamics (B2). Roles stay in their own lane below.

Role 1 — General Carbon Theory (lead; carries the heaviest load)

Mandate. Turn the headline from "conditional for general F" into a proved, finite-optimum, general-distribution theorem — the paper's title depends on it.

- **Existence & uniqueness.** Prove coercivity of profit Π and a primitive condition on F (e.g. log-concave density) under which any interior critical point is the unique global maximizer.
- **Primitive shift-ratio condition.** Replace the endogenous-window "$g(x) = F_{K_R}/f$ non-increasing" with a support-wide condition on F that *implies* it, and prove the implication.
- **General-F non-emptiness (keystone).** Under that primitive plus a tail/skew condition, prove carbon *falls somewhere* in C_R for any such F — upgrading the current conjecture to a theorem.
- **Finite-optimum carbon win.** Give a primitive condition under which carbon strictly falls at a *finite, interior* K_R^* (not the $K_R^*{=}\infty$ corner).
- **Main theorem + re-scope.** State the headline theorem, align the abstract/title to exactly what is proved, and write the economic intuition at journal quality.

Role 2 — Solver & Numerical Methods

Mandate. One coherent, reproducible numerical backbone demonstrating the theory at a finite interior optimum.

- **One carbon-sign narrative.** Replace the scattered "2/32" counts with a principled sweep over Role 1's primitive condition on a curated family, showing carbon falls at a finite interior optimum.
- **Hessian certificate.** A reproducible check that $\mathcal{H} \prec 0$ on the interior path for Role 1's distribution class (backstops existence).
- **Analytic-cofactor solver.** Replace finite-difference cofactors with analytic ones (removes the noisy knife-edge points); document tolerances.
- **One-command reproducibility.** A single `reproduce` harness covering the 15 exact-formula checks plus the carbon sweep.

Role 3 — Comparative Statics & Robustness

Mandate. Close the cross-distribution robustness layer Week 1 left at "fixed capacities," and deliver the sharpness example.

- **Cross-F least-favorable.** Sign the capacity covariation dK^*/dF along the shape class $\mathcal{U}$, turning the fixed-capacity result into a genuine least-favorable-distribution theorem.
- **Robust certificate.** Make $C_R \le C_R^\dagger \Rightarrow \Delta \ge 0$ hold for *every* $F \in \mathcal{U}$, or prove the obstruction cleanly.
- **Reversed-channel counterexample.** Construct a realized F with g increasing $\Rightarrow K_R^* < K_R^\circ$ — the sharpness example Week 1 could not produce.
- **Repair the limit.** Replace the $\mathbb{E}[\text{fossil}^*] \to 0$ a.s.-corner step with a dominated-convergence / finite-K_R argument.

Role 4 — Theory Hardening, Regimes & Positioning

Mandate. Make the four pieces one manuscript a referee reads as a single, consistent paper.

- **Unify the manuscript.** One Assumption 1 ($R > c + C_M$) and one carbon decomposition (Role 1's two-force split) everywhere; one notation table.
- **Complete the regime map.** Fold Role 1's finite-optimum statement into the small-FCL / interior / full-coverage map plus the C_R boundedness axis, as one consistent theorem.
- **Theory positioning.** Write the related-work section and bibliography on the *methods* A1 uses — monotone comparative statics (Topkis/Milgrom–Shannon), interruptible/demand-response economics, and distributionally-robust / least-favorable-distribution analysis.
- **Assemble & edit.** Move all "verified to 10^{-k}" out of proof bodies into the numerical section; align every claim with its theorem.

Sequencing

All four start now and share only notation + the Week-1 closed forms. Role 1 delivers two primitives early — the primitive shift-ratio condition (unblocks Role 2's sweep and Role 3's class $\mathcal{U}$) and the existence condition (unblocks Role 3's covariation and Role 2's Hessian check). Roles 2–4 hand Role 4 their final statements for assembly. **Acceptance gate:** a referee cannot reject on rigor once Role 1 (keystone) and Role 4 (consistency + positioning) land; Roles 2–3 turn acceptance into a strong one.